USER GUIDE · VIXL 0.26.0



A field guide for humans

How to make, check and ship editable designs with Vixl: the install, the first design, the working loop and everything that comes after.

IDEAS BECOME EDITABLE.

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HOW TO READ THIS GUIDE

Read 01–04 once, in order. After that, dip in. Commands are shown as you'd type them in a terminal; lines starting with **#** are comments. Everything here works offline except font downloads and AI providers.

01

What Vixl is

Ask an AI for a poster and you usually get a *picture* of a poster. Vixl gives you the real thing: layers, live text, fonts, a history you can rewind, and a master you can still change next week.

Programmable

One JSON operation format behind the CLI, Python, MCP and REST. Batches of up to 10,000 operations apply atomically, so a half-finished edit never lands.

Editable

Text stays text, effects stay a stack and history branches. Charts open in PowerPoint as charts; forms export as PDFs people can fill in.

Checked

Contrast, spacing, bounds, fonts, trim and bleed are measured before export, so nobody finds out at the print shop.

WHAT YOU GET BACK

- A **.vixl master**: one portable file holding layers, pages, fonts and history.
- **Exports** in the format the job needs: PNG, PDF, SVG, PPTX, video and more.
- **A check report** that says what was measured and what passed.

02

Install

Pick the route that matches how you work. Both give you the same `vixl` command, the same file format and the same results.

WINDOWS

The installer

Download **Vixl-Setup-VERSION-windows-x64.exe** from GitHub Releases. It bundles Python, REST and MCP, installs for your account and updates itself with verified downloads.

MACOS · LINUX · WINDOWS

Python 3.11+

Install from a source checkout into a virtual environment. Add **server** for REST and live review, **pdf** for importing PDF pages. MCP is always included.

terminal

```
# from a source checkout
$ python -m venv .venv
$ source .venv/bin/activate
$ python -m pip install -e '.[server,pdf]'

# confirm, then keep it current
$ vixl --version
$ vixl update --check
$ vixl update
```

vixl not found? Reopen the terminal, or activate the virtual environment you installed into. Pip installs update with pip; the Windows build updates itself.

03

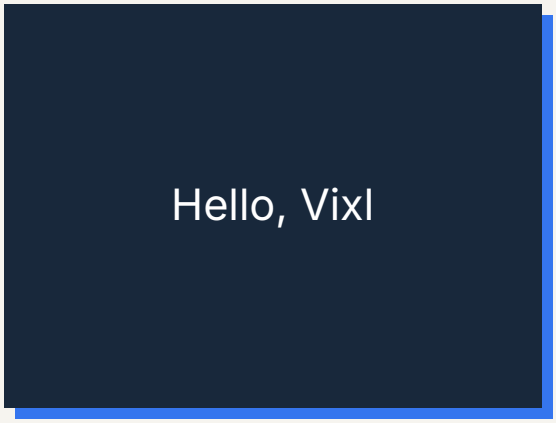
Your first design

Six commands take you from nothing to a checked, exported image that is still fully editable.

hello.vixl

```
$ vixl new 800x600 --background '#18283b' -o hello.vixl
$ vixl -p hello.vixl text add 'Hello, Vixl' --name title \
      --size 64 --color '#ffffff'
$ vixl -p hello.vixl align title center
$ vixl -p hello.vixl check --checks bounds contrast
$ vixl -p hello.vixl render --out hello-preview.png
$ vixl -p hello.vixl export hello.png
```

THE RESULT



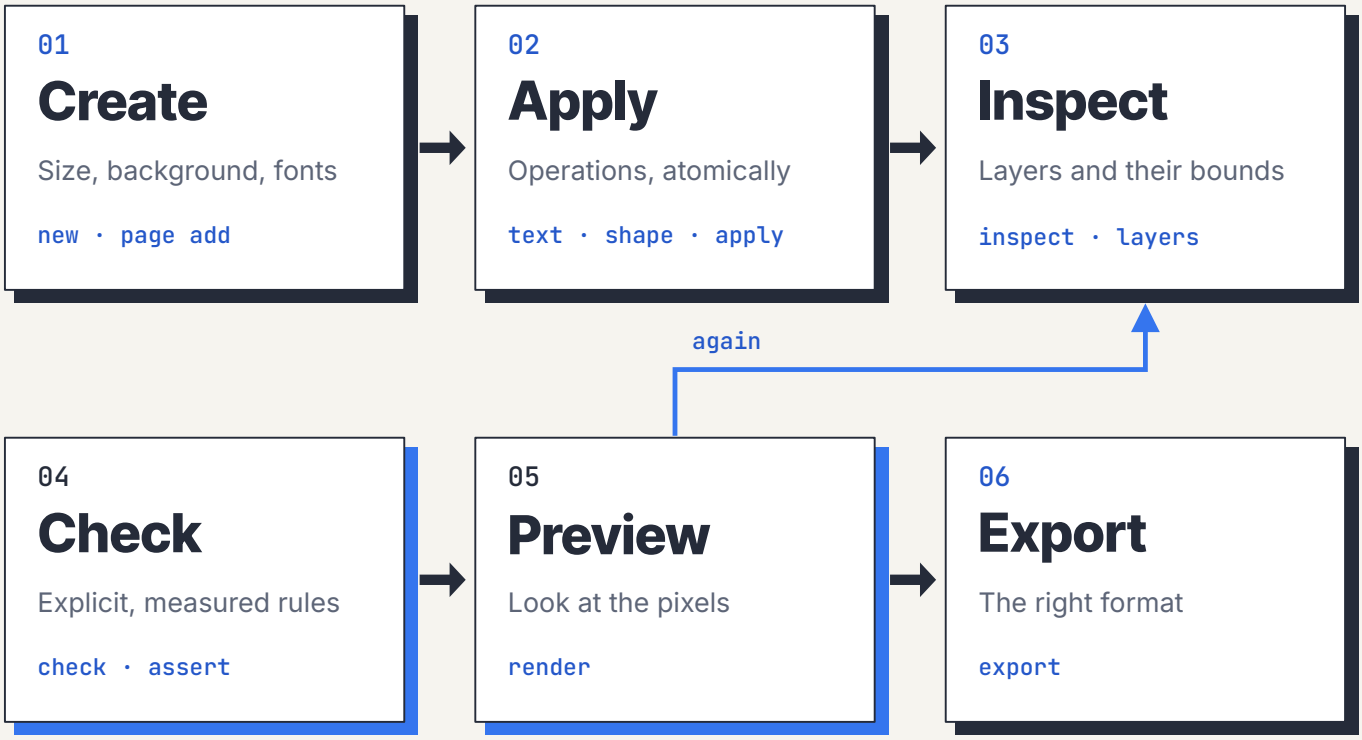
1. **new** makes an 800 × 600 document with a background.
2. **text add** puts a real text layer on it, named *title*.
3. **align** centres it. Names let you address layers later.
4. **check** measures bounds and contrast and reports back.
5. **render** previews; **export** writes the PNG. Look before you hand it over.

Editing commands save as they go. Changed your mind? `vixl -p hello.vixl undo`.

04

The working loop

Every piece follows the same six steps. The last three repeat until the checks pass and the preview looks right.



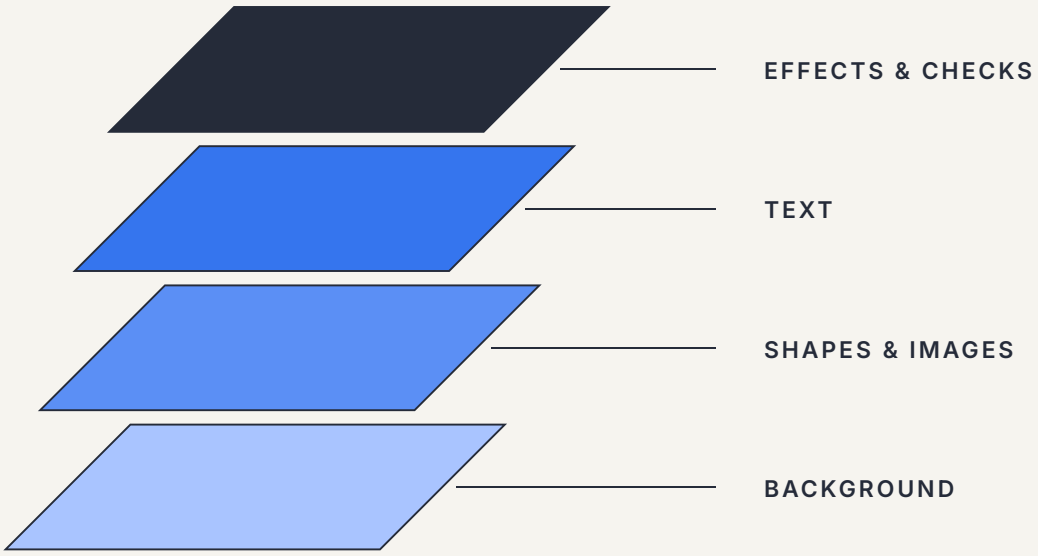
NOT RIGHT YET?

Revise, then inspect, check and preview again. `undo` and `checkpoint` make experiments cheap: nothing you try is permanent.

05

Inside a .vixl file

A Vixl document is one portable file. Fonts and images travel inside it, so it renders the same on any machine.



Layers

Every element is a named layer with a stable ID: text, shapes, paths, images, groups, paint, charts, fields.

Pages & masters

Decks, booklets and carousels. Masters hold what repeats, like the header on this page.

Variables

Write `${headline}` once, then export a hundred versions with different copy.

History

Undo, redo, named checkpoints and branches. Try something bold; come back.

06

Four ways in

The command line, Python, an AI agent over MCP or any HTTP client: they all speak the same JSON operations, so what works in one works in all four.

CLI

FOR YOU, IN A TERMINAL

```
$ vixl -p poster.vixl text add 'Sale'
$ vixl -p poster.vixl undo
```

Python

FOR SCRIPTS AND NOTEBOOKS

```
p = Project.load('poster.vixl')
p.apply([{'type': 'opacity',
  'target': 'logo', 'value': 0.7}])
p.save()
```

MCP

FOR AI AGENTS

```
$ vixl mcp --workspace . \
    --tools core --schema slim
```

REST

FOR ANY HTTP CLIENT

```
$ vixl -p poster.vixl serve
$ curl 127.0.0.1:8765/render \
    -o preview.png
```

Every operation has a schema: `vixl schema text` lists its fields, and `--dry-run` shows what a batch would change without changing anything.

07

Working with an agent

Connect an agent over MCP and it does the clicking. Your job is the brief. Good briefs name the purpose, the place and the proof.

A BRIEF THAT WORKS

"A 1080 × 1350 Instagram post for our Friday open studio. Headline *OPEN STUDIO*, date *Friday 19:00*. Use our brand.json. Dark background, one accent. Check contrast and safe area, show me a preview, then export PNG and keep the master."

Say what it's for

Poster, slide, form, sticker. The purpose picks sensible defaults.

Name the size and place

A preset like *instagram-post* or *letter*, and where it will be seen.

Ask for a preview

Look before export. Request changes by layer name.

Give the real words

Placeholder copy hides layout problems until it's too late.

Ask for checks

Contrast, bounds, safe area, print. Ask to see the report.

Keep the master

The .vixl file is the design. Exports are copies.

08

What you can make

One engine, a lot of outputs. Each of these has its own guide in the docs folder.

Posters & social
docs/sizes-and-layouts.md

Slides & decks
docs/slides.md

Fillable forms
docs/forms.md

Charts & tables
docs/charts.md

Diagrams
docs/diagrams.md

Organic & painted
docs/organic.md

Motion & video
docs/animation-authoring.md

Logos & icons
docs/production.md

Emoji packs
docs/emojis.md

Photo filters
docs/artistic-filters.md

Brands & templates
docs/brands.md

Print with bleed
docs/color-and-print.md

09

Check before you ship

A check is a measurement, not an opinion. Ask for the ones that matter for this piece and read what comes back.

```
$ vixl -p poster.vixl check --checks bounds contrast safe_area
$ vixl -p deck.vixl check --checks deck --pages all
```

`bounds`	Nothing important falls off the canvas
`contrast`	Text is readable against what's behind it
`safe_area`	Key content clears platform overlays and trims
`fonts`	Every glyph comes from a real, embedded font
`min_font`	No text is too small for the output size
`print`	Ink limits, resolution, trim and bleed
`color_vision`	Meaning survives colour-vision differences
`accessibility`	Alt text, reading order and roles
`deck`	Every page of a deck, plus deck-wide rules
`placeholders`	No leftover [Label] or lorem ipsum

Thirty-four checks in all; `vixl check --help` lists them. A passing check covers the rule you asked for, not every possible problem, so still look at the preview.

10

Export

Choose the format by where the work is going. Exports never overwrite an existing file unless you say so.

FORMAT	FOR	WHAT SURVIVES
PNG · WebP	Screens and the web	Transparency, any size
JPEG · TIFF	Photos and print rasters	CMYK with your printer's ICC profile
SVG	Scalable web graphics	Shapes, text and gradients as vectors
PDF	Print and documents	Vector pages, real text, trim and bleed
Fillable PDF	Forms people fill in	Real form fields over the artwork
PPTX	Presentations	Editable text, charts and speaker notes
HTML	Present from a browser	One file with a speaker view
PSD	Handoff to Photoshop	Named pixel layers and groups
WebP · GIF · MP4	Motion	Timelines; MP4 needs ffmpeg

```
$ vixl -p deck.vixl export deck.pdf
$ vixl -p deck.vixl export deck.pptx
$ vixl -p post.vixl export post.png --pages all
```

11

When things go wrong

Start with `vixl --version` and the exact error. Errors are structured: they name the field and usually suggest the fix.

⚠ **vixl is not found**	Reopen the terminal or activate the environment you installed into.
⚠ **A command edits the wrong file**	Always pass `-p file.vixl`. MCP tools take `document=`.
⚠ **Unknown operation or field**	Check `vixl schema NAME` and follow the error's suggestions.
⚠ **Layer not found**	Inspect the layers and pages; use the stable ID if names change.
⚠ **Text is cut off or tiny**	Inspect its bounds, give it a bigger box, review at real size.
⚠ **Export refuses the file**	It won't overwrite. Pick a fresh name or pass overwrite.
⚠ **PDF import or serve missing**	Install the `pdf` or `server` extra in the same environment.

```
# get a design back
$ vixl -p campaign.vixl undo
$ vixl -p campaign.vixl checkout approved-layout
```

Cheat sheet

Add `-p file.vixl` to each. `vixl COMMAND --help` explains any of them.

DOCUMENTS

`new 1080x1350 -o a.vixl`
Start a document

`inspect`
What's in it

`pages`
List pages and masters

MAKE

`text add 'Hi' --name t`
Add text

`shape rectangle --name r`
Add a shape

`chart · table · diagram`
Data and structure

ARRANGE

`align t center`
Align a layer

`move t --x 40 --y 40`
Move it

`group g a b`
Group a and b as g

HISTORY

`undo · redo`
Step back and forward

`checkpoint v1`
Name this moment

`checkout v1`
Return to it

REVIEW

`check --checks contrast`
Measure

`render --out p.png`
Preview

`view`
Live review in a browser

SHIP

`export out.pdf`
Write a file

`export s.png --pages all`
One image per page

`mcp --workspace .`
Hand it to an agent



Ideas become editable.

github.com/jxburros/Vixl · docs/README.md

ABOUT THIS GUIDE

This booklet is a Vixl document: every word is a live text layer, every mark a path. It was built by `examples/build_user_guide.py`, checked for bounds, contrast and fonts, and exported to PDF. Set in Inter Tight, Inter and JetBrains Mono.